Computer Science & IT

COMPILER DESIGN
(CD)

Parsers

Lecture No. 8



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Recap of Previous Lecture



- Topic
- Topic
- Topic

LLC1)

Topics to be Covered



- Topic
- Topic
- Topic

LL(1)
LR( )

Extra Ordinary Individuals: Stories to Ignite Student Motivation

Dr. Anil Sharma led a CSIR lab at IISc focusing on bio-pesticide formulations. When small-scale tests yielded inconsistent results, he partnered with village cooperatives for on-farm trials. Local feedback drove iterative formulation tweaks under field conditions. His final bio-pesticide blend cut crop losses by 40% for smallholders, reducing chemical runoff and empowering farmers with sustainable tools.

Lesson: Embedding research within community trials ensures innovations solve real-world problems effectively.



Extra Ordinary Individuals: Stories to Ignite Student Motivation

Prof. Jayant Narlika: Renowned for his work on cosmology at IISc, returned from Cambridge to establish India's first black-hole research group. Lacking supercomputers, he devised analytical models and trained students in pencil-and-paper calculations. When funding for international conferences was scarce, he published groundbreaking papers locally, attracting global collaborators. His mentorship created a generation of theoretical astrophysicists who now lead institutes worldwide.

Lesson: Visionary academic leadership can germinate world-class research even without lavish resources.

Extra Ordinary Individuals: Stories to Ignite Student Motivation

H. S. Shankar returned to IIT Madras as department head determined to overhaul outdated curricula. He surveyed industry needs and student aspirations, then formed an alumni-faculty council for continuous feedback. When bureaucratic red tape stalled approvals, he demonstrated pilot courses, securing rapid endorsements. His reforms aligned syllabi with cutting-edge research and startup culture, raising placement rates by 30%.

Lesson: Engaging stakeholders through prototypes accelerates institutional change and relevance.

Extra Ordinary Individuals: Stories to Ignite Student Motivation

R. B. Bapat : As dean at IIT Bombay, R. B. Bapat noticed students struggled with remote learning. He launched interactive virtual labs using low-bandwidth video tools, training faculty to deliver hands-on modules online. When initial uptake lagged, he organized peer-mentor sessions and open-source tutorial drives. Within a semester, remote student engagement doubled, and virtual labs became a model for other campuses.

Lesson: Adopting accessible technology and peer support can rejuvenate education delivery at scale.

	first	follow
$S \to A\ X$	{a, b, c, d}	{$}
$A \to Bb / Cd$	{a, b, c, d}	{$}
$B \to aB / \epsilon$	{a, ε}	{b}
$C \to cC / \epsilon$	{c, ε}	{d}

LL(1)

	a	b	c	d	$
S	S → A	S → A	S → A	S → A	
A	A → Bb	A → Bb	A → Cd	A → Cd	
B	B → aB	B → ε			
C			C → c	C → ε	

$S \rightarrow \underline{aA}\overset{\{a\}}{\uparrow}\underline{a} / \epsilon \quad \{a\}$

$A \rightarrow abS / \epsilon$

	First	follow
$S \rightarrow aAa / \epsilon$	$\{a, \epsilon\}$	$\{\$, a\}$
$A \rightarrow abS / \epsilon$		

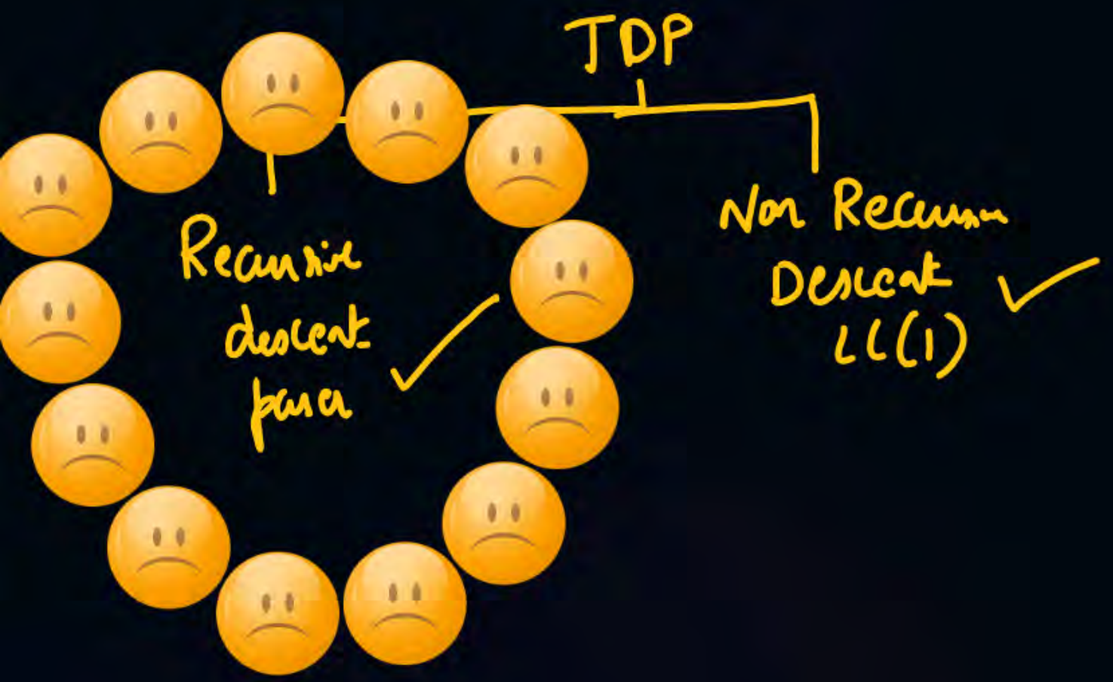
Not (((

	a	b	$
S	$S \rightarrow aAa$ / $S \rightarrow \epsilon$		

$S \rightarrow iEtSS' / a$

$S' \rightarrow eS / \epsilon$ ✗ (LL(1))

$E \rightarrow b$


TDP
Recursive descent parser
Non Recursive Descent LL(1)

Recursive descent parser.-

$E \rightarrow i E'$

$E' \rightarrow +iE' / \epsilon$

```
E()
{ if(l == i)
  { match(i)
    E'()
  }
}
```

```
E'()
{ if(l == '+')
  { match('+');
    match('i');
    E'()
  }
else
  ε
}
```

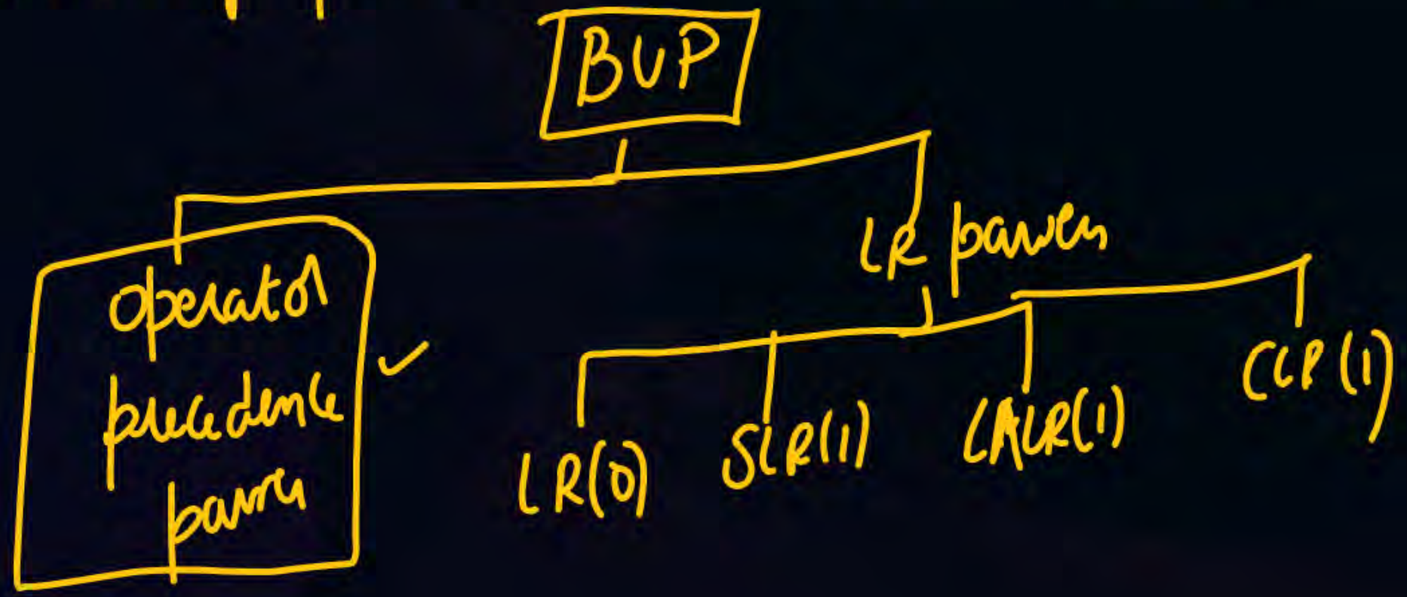
```
char l;
l = getchar();
match(char t)
{ if(l == t)
    l = getchar();
  else
    Printf("Error");
}
```

```
main
{ E()
  if(l == $)
  Printf("Parsing Success")
```

Bottom up parsers:

It starts construction of trees starting at leaves and working up towards the root. It uses Reverse of RMD

Bottom up parsers are also called SR parsers



Operator precedence parser

THANK - YOU